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Task:

We have birth date (month) data from N people. Let's count how many of them have birthday during the winter!

Specification:

Input: $n \in \mathbb{N}$, $m \in \mathbb{N}[1..n]$

Output: $\text{cnt} \in \mathbb{Z}$

Precondition: $\forall i \in [1..n]: (1 \leq m[i] \leq 12)$

Postcondition: $\text{cnt} = \text{COUNT}(i=1..n, m[i] < 3 \text{ or } m[i] = 12)$

Pattern: counting

Algorithm:

Pattern		Task
$b..e$	$\sim$	$1..n$
cnt	$\sim$	cnt
$A(i)$	$\sim$	$m[i] < 3 \text{ or } m[i] = 12$

$\text{cnt} := 0$	
$i = 1..n$	
T	$m[i] < 3 \text{ or } m[i] = 12$
	F
$\text{cnt} := \text{cnt} + 1$	-

$\text{cnt} := 0$	
$i = b..e$	
T	$A(i)$
	F
$\text{cnt} := \text{cnt} + 1$	-

Test cases:

1. invalid input: $n: 2$; $m: [-2, 4]$ -> invalid...
2. nobody has birthday during the winter: $n: 3$; $m: [5, 6, 5]$ -> $\text{cnt}=0$
3. 1 person has... and he is not the first or last one: $n: 3$; $m: [5, 1, 5]$ -> $\text{cnt}=1$
4. 3 (more) people have...: $n: 5$; $m: [12, 5, 1, 5, 7]$ -> $\text{cnt}=2$
5. all people have...: $n: 3$; $m: [12, 1, 2]$ -> $\text{cnt}=3$